Sang Nguyen

San Francisco Bay Area | 510-424-2625 | sangnguyen.work | sangoscar99

Experience

Blackhawk Network July 2022 – Present
Software Engineer II *Pleasanton, CA*

- Architected and delivered a cloud-native Order Management platform processing **100K+ orders per day**, leading end-to-end system design across distributed services, asynchronous workflows, data persistence, caching, and AWS infrastructure to improve reliability and scalability.
- Containerized backend services with **Docker** and deployed them to **AWS ECS** through **Jenkins** CI/CD pipelines within **CloudFormation**-provisioned environments, enabling reliable and repeatable service releases.
- Engineered a reusable **Java** query translation library that converts **RSQL** expressions into native **MongoDB Atlas Search** pipelines, packaging versioned artifacts for integration across downstream services and standardizing complex search capabilities.
- Engineered event-driven architecture using **AWS EventBridge**, **SNS**, and **SQS** to asynchronously propagate order-notification events across downstream services, aligned with order lifecycle state transitions.
- Championed **TDD/BDD** practices using **JUnit**, **Mockito**, **Cucumber**, and **Karate Framework**, achieving a **40% increase** in code coverage, **50% reduction** in production bugs, and **zero critical vulnerabilities**.
- Served as a go-to engineer for on-call production support, leading root-cause investigations across distributed services, databases, queues, and application logs to drive rapid remediation; proactively identified defects in non-production validation before customer impact.

Contra Costa College Feb 2018 – Dec 2019
Peer Tutor *San Pablo, CA*

- Tutored **60+ students** in **Computer Science, Mathematics**, and **Statistics**, achieving a **95% grade improvement rate** through personalized instruction and progress monitoring.

Education

University of California, Davis July 2020 – May 2022
Bachelor of Science in Computer Science | *GPA: 3.67 / 4.00* *Davis, CA*

Contra Costa College July 2017 – May 2020
Associate in Arts, Math & Science | *GPA: 3.90* | *Dean's List, President's Scholar* *San Pablo, CA*

Projects

Tracitivity | *Java, D3.js, OAuth, REST API, Google API* Mar 2021 – May 2021

- Developed **OAuth 2.0** authentication to securely exchange access tokens with protected **Google APIs**, enabling safe user authorization.
- Built a **full-stack web application** for storing and retrieving activity sessions, featuring interactive data visualizations via **D3.js** and an automated **reminder system**.

Shop Till You Drop | *React.js, REST API, GitHub Actions, CI/CD* May 2021 – June 2021

- Built a **responsive** web app supporting desktop, tablet, and mobile using reusable, accessible **React** components.
- Optimized **College Scorecard API** calls to reduce overhead and help students estimate tuition costs, managed via **CI/CD** pipeline with **GitHub Actions**.

Technical Skills

Languages: Java, Python, JavaScript, SQL, HTML/CSS
Technologies: Spring Boot, Spring Framework, Docker, AWS ECS, ECR, EventBridge, SNS, SQS, CloudFormation, MongoDB, MongoDB Atlas Search, Redis, Maven, Jenkins, JUnit, Mockito, Cucumber, Karate, Git, Bash
Specialties: Distributed Systems, Microservices Architecture, Event-Driven Architecture, REST API Design, AWS, NoSQL Databases, Database Optimization, Caching Strategies, Test-Driven Development